

Topic 2.11 - Logarithmic Functions

Practice Worksheet

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. For contextual models, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Describe logarithmic domain, range, asymptotes, monotonicity, concavity, and end behavior.
- Connect equal multiplicative changes in input to equal additive changes in output.
- Analyze and construct transformed logarithmic functions with explicit restrictions.

Section A: Foundation and representation

Question 1

Core skill

No calculator

Analyze $f(x) = \log_5 x$: state domain, range, intercepts, asymptote, monotonicity, concavity, and end behavior.

Question 2

Core skill

No calculator

For $g(x) = -2 \log_3(x + 1) + 4$, state the domain, asymptote, anchor point, monotonicity, and concavity.

Question 3

Core skill

No calculator

Find the x -intercept of $h(x) = 3 \log_2(x - 4) - 6$.

Question 4

Core skill

No calculator

Find the domain and asymptote of $p(x) = \ln(2 - 3x) + 5$.

Question 5

Core skill

No calculator

Construct a function of the form $f(x) = a \log_4(x - 1) + k$ that passes through $(2, 3)$ and $(5, 7)$.

Question 6

Comparison

No calculator

Compare $\log_2 x$ and $\log_8 x$ for $x > 1$ and for $0 < x < 1$.

Section B: Application, comparison, and error analysis

Question 7

Core skill

No calculator

For $f(x) = 5 \log_3 x + 2$, find $f(27x) - f(x)$.

Question 8

Multiplicative change

No calculator

A logarithmic model $L(x) = 12 + 7 \log_2 x$ is defined for $x > 0$. By what factor must x be multiplied to increase L by 21?

Question 9

Error analysis

No calculator

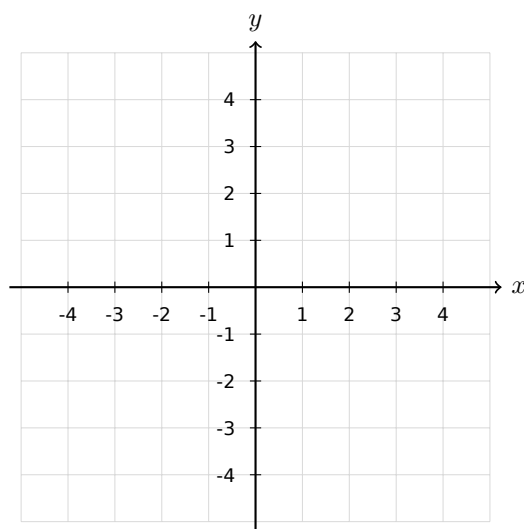
A student states the domain of $\log(x^2 - 4)$ is $x \neq \pm 2$. Correct the error.

Question 10

Graphing

No calculator

Sketch $q(x) = \log_2(x + 2) - 1$, labeling the asymptote, anchor point, and x -intercept.



Question 11

Parameter reasoning

No calculator

Determine all values of a for which $f(x) = a \ln x$ is decreasing and concave up.

Question 12

Context

No calculator

A model is $S(x) = 30 + 8 \log_{10}(x/5)$ for $x > 0$. Interpret the change in S when x is multiplied by 100.

Section C: AP-style multiple choice

MCQ 1

No calculator

The vertical asymptote of $f(x) = \log_3(x - 7) + 2$ is:

- A. $x = -7$
- B. $x = 7$
- C. $y = 2$
- D. $y = 7$

MCQ 2

No calculator

Which function is decreasing and concave up on its domain?

- A. $\log_2 x$
- B. $-\log_2 x$
- C. 2^x
- D. -2^x

MCQ 3

No calculator

The anchor point of $4\log_5(x + 3) - 2$ is:

- A. $(-3, -2)$
- B. $(-2, -2)$
- C. $(1, 2)$
- D. $(3, -2)$

MCQ 4

No calculator

For $f(x) = 3\log_2 x$, what is $f(4x) - f(x)$?

- A. 2
- B. 3
- C. 6
- D. 12

Section D: Free-response reasoning

FRQ 1 - Complete logarithmic analysis

No calculator

6 points

Let $f(x) = -4\log_3(x-2)+6$. (a) State domain, range, vertical asymptote, and anchor point. (b) Find the x -intercept. (c) Describe monotonicity, concavity, and end behavior. (d) Find $f(9(x-2)+2)-f(x)$ for $x > 2$.

FRQ 2 - Construct and interpret

No calculator

6 points

A logarithmic model has vertical asymptote $x = 5$. It passes through $(6, 2)$ and $(14, 8)$, and uses base 3. (a) Construct the model in the form $a\log_3(x-5)+k$. (b) State its domain and monotonicity. (c) Find the input at which the model output is 5. (d) Explain how the output changes when $x - 5$ is multiplied by 27.
